

EXPERIMENT 1

THEORY:

- a) Creation and loading different types of datasets in Python using the required libraries.
 - i. Creation using pandas
 - ii. Loading CSV dataset files using Pandas
 - iii. Loading datasets using sklearn
- b) Write a python program to compute Mean, Median, Mode, Variance, Standard Deviation using Datasets
- c) Demonstrate various data pre-processing techniques for a given dataset. Write a python program to compute
 1. Reshaping the data,
 2. Filtering the data,
 3. Merging the data
 4. Handling the missing values in datasets
 5. Feature Normalization: Min-max normalization

CODE:

```
import pandas as pd

df = pd.read_csv("data.csv")

print(df.head())
```

	NOC	ISO3	Year	population	gdp_per_capita	income_group	host_country	\
0	AFG	AFG	2004	23560654.0	221.763654	Low income		0
1	AFG	AFG	2008	26482622.0	381.733238	Low income		0
2	AFG	AFG	2012	30560034.0	651.417134	Low income		0
3	AFG	AFG	2016	34700612.0	522.082216	Low income		0
4	ALB	ALB	1992	3247039.0	200.852220	Low income		0
				athletes_sent	sports_participated	events_participated	\	
0				5	4	5		
1				4	2	4		
2				6	4	6		
3				3	2	3		
4				7	4	8		
				female_athlete_percentage	prev_total_medals	prev_medals_per_athlete	\	
0				40.000000	0.0	0.000000		
1				25.000000	0.0	0.000000		
2				16.666667	1.0	0.250000		
3				33.333333	1.0	0.166667		
4				22.222222	0.0	0.000000		
	Gold	Silver	Bronze	total_medals	medals_per_athlete			
0	0.0	0.0	0.0	0.0	0.000000			
1	0.0	0.0	1.0	1.0	0.250000			
2	0.0	0.0	1.0	1.0	0.166667			
3	0.0	0.0	0.0	0.0	0.000000			
4	0.0	0.0	0.0	0.0	0.000000			

```
import numpy as np
```

```
data = np.array(df.values)
```

```
reshaped_data = data.reshape(data.shape[0], data.shape[1])
```

```
print(reshaped_data)
```

```
[['AFG' 'AFG' 2004 ... 0.0 0.0 0.0]
 ['AFG' 'AFG' 2008 ... 1.0 1.0 0.25]
 ['AFG' 'AFG' 2012 ... 1.0 1.0 0.16666666666666666]
 ...
 ['ALG' 'DZA' 2008 ... 1.0 2.0 0.0357142857142857]
 ['ALG' 'DZA' 2012 ... 0.0 1.0 0.0263157894736842]
 ['ALG' 'DZA' 2016 ... 0.0 2.0 0.03125]]
```

```
filtered_df = df[
```

```
    (df['Year'] >= 2012) &
```

```
    (df['income_group'].isin(['Low income', 'Lower-middle income']))
```

```
]
```

```
print(f"Number of rows after filtering: {len(filtered_df)}")
```

```
Number of rows after filtering: 19
```

```
iso_mapping = pd.DataFrame({
```

```
    'ISO3': df['ISO3'].unique(),
```

```
    'Region': ['Region_' + str(i) for i in range(len(df['ISO3'].unique()))]
```

```
})
```

```
merged_df = pd.merge(df, iso_mapping, on='ISO3', how='left')
```

```
print(merged_df[['NOC', 'ISO3', 'Region']].head())
```

```
   NOC ISO3  Region
0  AFG  AFG  Region_0
1  AFG  AFG  Region_0
2  AFG  AFG  Region_0
3  AFG  AFG  Region_0
4  ALB  ALB  Region_1
```

```
print(df.isnull().sum())
```

```
df_cleaned = df.copy()
```

```
df_cleaned['gdp_per_capita'] =
df_cleaned['gdp_per_capita'].fillna(df_cleaned['gdp_per_capita'].mean())
df_cleaned['income_group'] = df_cleaned['income_group'].fillna('Unknown')
```

```

NOC                0
ISO3               0
Year              0
population         0
gdp_per_capita     0
income_group       0
host_country       0
athletes_sent      0
sports_participated 0
events_participated 0
female_athlete_percentage 0
prev_total_medals  0
prev_medals_per_athlete 0
Gold              0
Silver            0
Bronze            0
total_medals      0
medals_per_athlete 0
dtype: int64
```

```
cols_to_normalize = ['population', 'gdp_per_capita']
df_normalized = df.copy()
for col in cols_to_normalize:
    col_min = df[col].min()
    col_max = df[col].max()
    df_normalized[col] = (df[col] - col_min) / (col_max - col_min)

print(df_normalized[['population', 'gdp_per_capita']].describe().loc[['min', 'max']])
```

```

      population  gdp_per_capita
min           0.0            0.0
max           1.0            1.0
```